paper_03_miniverse_cosmology

Paper 3 Brief: Mini-Verse Sandboxes as Configurable Cosmological Substrates

Authority state: `INTERNAL_RESEARCH`

Claim boundary: `C3`, `C4`, `C8`, `C9`, `C10`; implementation claims held as `evidence_needed_before_release` because matching mini-verse files were not located in the inspected repo surface.

Abstract

This paper turns the mini-verse material in the canonical packet into a standalone preprint on configurable cosmological sandboxes for autonomous-intelligence experiments. It treats each mini-verse as an isolated universe instance with its own constants, topology, particle population, cosmological evolution, entropy accounting, and bounded inter-verse communication channels. The paper's contribution is a sandbox architecture for varying physical assumptions, not a proof that agent intelligence improves under those assumptions. Implementation and experimental claims should remain bounded until reproducible runs compare agent behavior across topologies and constants.

Distinct Thesis

Configurable mini-verses can serve as experimental substrates for studying how autonomous-agent behavior changes under different physical constants, topological constraints, cosmological phases, and inter-domain boundary conditions.

Contribution Surface

- Mini-verse object model: constants, topology, geometry, particles, and evolution state.
- Topology catalog: flat infinite, flat periodic, spherical, hyperbolic, de Sitter, and anti-de Sitter.
- Friedmann evolution as universe-level simulation policy.
- Thermodynamic arrow of time using particle and horizon entropy accounting.
- Inter-verse boundary channels: wormholes, entanglement-like information channels, and membranes.

Bounded Claims

Claim	Class	Proof posture	Public wording
Mini-verses can be instantiated with configurable physical constants and topologies.	Potential `C1` / `C8`	Not located in inspected repo surface.	"The research architecture defines mini-verse instances with configurable constants and topology choices; implementation claims should cite matching repository files once located."
Friedmann equations govern cosmological evolution inside each mini-verse.	`C10` / potential `C1`	Equations are standard; implementation needs evidence.	"The proposed evolution model uses Friedmann-style equations for scale-factor dynamics."
Entropy tracking provides an arrow of time.	`C3` / `C9`	Needs model specification and validation.	"Entropy accounting is proposed as a runtime indicator of thermodynamic directionality."
Inter-verse channels enforce causality through bandwidth limits.	`C3` / `C8`	Needs formal semantics and tests.	"Inter-verse channels are specified with bandwidth limits intended to constrain transfer."

Proposed Outline

1. Introduction: why isolate cosmological sandboxing from engine internals.
2. Mini-verse object model and lifecycle.
3. Topology catalog and boundary behavior.
4. Friedmann evolution and phase classification.
5. Entropy accounting and thermodynamic direction.
6. Inter-verse connection types and boundary constraints.
7. Experimental designs for agent behavior under varied constants/topologies.
8. Validation requirements and failure modes.
9. Relationship to physics, geometry, and agent architecture papers.

Required Evidence Before Preprint Release

- Formal mini-verse schema or class interface.
- Reproducible examples for at least three topologies.
- Scale-factor evolution traces for radiation, matter, dark-energy, and contracting phases.
- Entropy tracking definition with numeric examples.
- Transfer tests showing bandwidth-limited inter-verse channels.

Minimized Overlap Rule

This paper should not re-derive Christoffel symbols or Velocity Verlet mechanics beyond short citations to Papers 1 and 2. Its novelty is the universe-level composition and experiment design layer.